

Thomas Simpson

Senior Software Engineer

thomasroyjob@gmail.com

linkedin.com/in/thomas-simpson-tech/

+1 713 597 6150

Mcallen, TX, 78504, USA

An expert Java software engineer with 11+ years specializing in back-end and microservices, as well as workflow/release engineering. Comfortable with a broad set of libraries/environments. Skilled in CI/CD and cloud deployment. Focused on test-driven, domain-driven, and object-oriented design principles.

Work Experience

Senior Consultant

Mar 2022 - Feb 2025

OSG

- Designed and developed cloud-native microservices using Java 11+, Spring Boot, and AWS Lambda, achieving 40% faster response times.
- Led the migration from monolithic to microservices architecture, using Docker, Kubernetes (EKS), and AWS ECS, reducing deployment time by 60%.
- Built and optimized RESTful APIs for high-traffic enterprise applications, handling 1M+ daily transactions with Spring Boot and AWS API Gateway.
- Integrated third-party services using Java 8 Streams and REST APIs, deploying the solution on AWS S3 for data storage.
- Enhanced data processing pipelines by implementing Kafka and AWS SQS, improving event-driven consistency and reducing latency by 35%.
- Managed CI/CD pipelines using Jenkins, GitHub Actions, and AWS CodePipeline, automating deployments and reducing manual intervention by 70%.
- Optimized system performance, identifying and addressing bottlenecks across both application code and cloud infrastructure.
- Designed and developed RESTful and SOAP-based web services, ensuring compliance with OpenAPI standards and utilizing Swagger for API documentation.
- Integrated and optimized API Gateways, including IBM DataPower and AWS API Gateway, to enhance security, performance, and scalability of microservices.
- Developed and maintained secure API endpoints with OAuth2 and JWT authentication, improving system integrity and user authentication flow.
- Implemented rate limiting, caching, and request validation in API Gateways, reducing API response time by 30%.

- Mentored junior developers, leading to a 20% increase in team productivity.

Senior Software Engineer

Feb 2018 - Mar 2022

Rootstack

- Developed scalable backend services using Java 8+, Spring Framework, and AWS RDS (PostgreSQL), improving database efficiency by 30%.
- Implemented OAuth2 and JWT-based authentication using Spring Security, securing cloud-based applications across multiple regions.
- Led the cloud migration of legacy applications to AWS and GCP, leveraging Terraform and CloudFormation for automated infrastructure provisioning.
- Built real-time analytics platforms using Apache Kafka and AWS Kinesis, processing over 10M events daily for data-driven insights.
- Delivered cloud-native solutions leveraging Java, Spring Boot, and AWS Aurora to create high-performance relational databases for enterprise systems.
- Created and consumed SOAP-based web services using JAX-WS, ensuring compatibility with legacy enterprise systems.
- Automated API testing and monitoring with Postman, JMeter, and Swagger, ensuring 99.9% uptime and reducing regression issues.
- Contributed to Java-based microservices and API development, integrating JavaScript-based front-end applications.
- Utilized GitHub Actions for CI/CD workflows, automating API deployments and improving version control efficiency.
- Conducted code reviews and best practice enforcement, reducing bug rates by 25%.

Software Engineer

May 2014 - Feb 2018

VironIT

- Designed and deployed Java-based microservices on AWS ECS and EKS, ensuring high availability with 99.99% uptime.
- Optimized database performance by refactoring SQL queries and indexing strategies, reducing query execution times by 40%.
- Integrated Redis and MongoDB for caching and NoSQL data handling, enhancing system throughput.
- Developed automated test suites using JUnit, Mockito, and Selenium, increasing test coverage to 85%.
- Coordinated with cross-functional teams to ensure the successful deployment of new features and enhancements in a fast-paced environment.

- Collaborated with DevOps teams to automate deployments using Docker, Kubernetes, and Terraform, reducing infrastructure provisioning time by 50%.

Projects

Event-Driven Order Processing System

Feb 2023

- Designed and implemented a real-time order processing system using Kafka, Spring Boot, and AWS Lambda, ensuring message ordering and fault tolerance. - Reduced order processing latency by 45%, improving the customer experience.

CI/CD Pipeline for Microservices Deployment

Feb 2022

- Developed an automated CI/CD pipeline using Jenkins, Docker, and Kubernetes, enabling seamless deployment of microservices across multiple environments. - Reduced deployment failures by 80% through rigorous testing and monitoring.

Cloud-Based Fraud Detection System

Jul 2020

- Built a fraud detection system leveraging AWS SageMaker and Python ML frameworks, identifying fraudulent transactions with 90% accuracy. - Integrated the system with Kafka for real-time streaming and anomaly detection.

Core Skills

Programming Languages: Java (Java 8+), C#, JavaScript, Shell Scripting (Python, Bash)

Frameworks and Libraries: Spring Boot, Spring Framework (Spring MVC, Spring Security, Spring Data), JPA/Hibernate, .NET, ReactJS / Angular, Spring Cloud

Cloud Technologies: AWS(Lambda, ECS, S3, RDS, DynamoDB, AuroraDB), Azure, GCP

Microservices and Event-Driven Systems: Microservices Architecture, Event Sourcing, Kafka

Databases: Relational Databases (Oracle, MySQL, PostgreSQL, Aurora), NoSQL Databases (MongoDB, Cassandra, Redis, DynamoDB, Elasticsearch), ORM (JPA, Hibernate)

DevOps & CI/CD: Jenkins, Git & GitHub, CI/CD Pipelines (Jenkins, GitHub Actions, AWS CodePipeline, Azure DevOps), Docker, Kubernetes, Terraform, SonarQube

Testing & Quality Assurance: JUnit / TestNG, Mockito, Cucumber, Selenium, Postman, Swagger

Agile & Collaboration: Agile (Scrum, Kanban), Jira (Issue Tracking), Confluence (Documentation)

Containerization & Orchestration:: Docker, Kubernetes, AWS ECS / EKS

Operating Systems: Linux / Unix (System Administration, Deployment), Windows (Basic System Interaction)

Education

University of Texas at Dallas

Aug 2010 - Aug 2014

Bachelor's Computer Science

GPA: 3.7/4.0

Awards

Employee of the Year

Jan 2022

Rootstack

Certificates

Oracle Certified Professional, Java SE 8 Programmer (OCPJP 8)

Oracle Corporation

AWS Certified Solutions Architect – Associate

Amazon Web Services